



DESN2000 SENG

Final Report

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1 Updated Software Requirements

This section logs the acceptance criteria that changed as a direct result of Sprint 1 and Sprint 2 prototype testing (Section 3.3 and Section 3.5). Each row links the original acceptance criterion, the testing feedback that prompted the change, the resulting modification, and the criterion’s validation status after retesting. The same changes are reflected as issue history on the team Jira board (project ST4).

Table 1. Requirements changelog: acceptance criteria updated as a result of user testing.

Requirement	Testing feedback	Original version	Modification	New version	Validation result
US-13 / AC-US13-04	In Iteration 1, four participants recovered independently, but only one did so within 60 seconds.	The system notifies the user of an empty search but does not name a recovery action.	Added a Broaden search action beside guidance that names which inputs can be changed to return results.	AC-US13-04 revised to require a visible recovery action and named changeable inputs, not just an empty-state notification.	Pass (Iteration 2: 5/5 independent, 44s median)
US-11 / AC-US11-02	Verification cues relying on an icon alone made verified-only filtering hard to confirm; users also could not distinguish identity verification from phone verification.	Only profiles with a verified badge appear under the verified-only filter; the badge is icon-only.	Paired the verification icon with readable status text and made the supporting tabs keyboard-operable (arrow, Home, End).	AC-US11-02 retained, with the verified-only filter result now required to expose readable status text alongside the icon.	Pass, but flagged: identity vs. phone verification confusion persisted into Iteration 2
US-12 / AC-US12-02	Three participants could not explain what the compatibility score measured; some read a high score as a safety signal rather than a preference match.	Only a percentage compatibility score was displayed.	Added a match-card explanation naming which saved preferences drove the score and which differences remain.	AC-US12-02 revised to require the detailed compatibility summary to state the evidence the score is based on.	Partial (Iteration 2: 3/5 correctly explained; treated as a persistent issue, Section 3.6)
US-09 / AC-US09-04	Participants sending a message with personal contact details received no clear confirmation that the details were blocked.	The system blocks a message containing contact details and directs the user to mutual contact sharing, with no explicit confirmation step.	The warning now blocks the content, explains the mutual-sharing condition, and announces the result through a live alert; the mobile view retains the conversation selector and New control.	AC-US09-04 retained, with the blocked-message result required to be explicitly announced to the user.	Pass (changed from Fail in Iteration 1 to Pass in Iteration 2)
US-10 / AC-US10-02	Participants could not tell whether a proposed rule had finished collecting acknowledgements or when it would activate.	The rule is added to the household rules list once the last acknowledgement is submitted, with no separate progress indicator.	Partial fixes applied to related household flows (dialog focus, roster validation, draft save confirmation); the approval-status indicator itself was not yet revised.	No revised acceptance-criteria text; the existing criterion was retained for Iteration 2 retesting.	Iteration 1: Fail; Iteration 2: Pass (AC text unchanged)

All five rows above are also being maintained as Jira issue history under project ST4 (<https://jira.st4.com>)

`//unsw-team-desn-seng2026.atlassian.net`), with each acceptance-criteria change linked to the corresponding test task (T1–T5) and sprint. AC-US10-02 passed in Iteration 2 without a text revision; the remaining prominence issue for the rule-approval status indicator remains open in Jira.

2 Accessibility Report

2.1 Rationale and Requirements Traceability

Accessibility was considered an essential part of the user experience rather than just a visual feature. In the interim report, we identified an international student persona who needs to evaluate unfamiliar accommodation listings and potential housemates despite having limited local knowledge and English proficiency. We also identified a domestic shared-living persona who relies on household agreements remaining clear and accessible after moving in. Both users depend on information that can be understood across different devices, languages, and interaction methods.

This was also reflected in our user testing. Participants included a user with colour vision deficiency, a mobile-first international student, and a participant whose first language was not English. Their feedback showed that if important information relies only on colour, keyboard navigation is interrupted, or translated content is difficult to understand, users may struggle to judge the trustworthiness of listings, communicate safely, or understand household agreements.

Two non-functional requirements formed the basis of the accessibility evaluation. NFR-03 requires all Sprint 1 screens (S-01 to S-06) to support keyboard navigation and meet WCAG 2.2 AA colour contrast requirements [1], while NFR-06 requires the household rules page to support at least five languages. The evaluation applies all 38 Level A and Level AA success criteria in the final WCAG 2.0 Recommendation [2].

Accessibility requirements were also linked directly to the project’s user stories and acceptance criteria. US-11 (AC-US11-01 and AC-US11-02) supports recognition of verified profiles, US-09 (AC-US09-04) ensures users can understand blocked contact-detail messages, US-02 (AC-US02-02) focuses on translated household rules, and US-10 (AC-US10-02 and AC-US10-04) supports creating, approving, and activating household rules. Linking accessibility to these user stories ensured that accessibility requirements were considered throughout the design process instead of being added at the end.

2.2 Evaluation Scope and Method

The W3C Template for Accessibility Evaluation Reports and the A11Y Project checklist guided the report structure and plain-language interpretation [3, 4].

The evaluation covered the six Sprint 1 prototype screens (S-01 to S-06), together with the login and notification dialogs that support these user tasks.¹ Although the Create Roster dialog was outside the Sprint 1 release scope, it reused the same dialog and form-validation components as Screens S-05 and S-06. It was therefore included as a component-level sample to assess the accessibility of shared interface elements.

The prototype was manually tested for keyboard navigation, focus management, DOM and ARIA attributes, language metadata, colour contrast, and responsive layouts at resolutions of 1470 × 726 and 390 × 844 pixels. Nine accessibility issues were identified. Screenshots were captured to document the interface state.

Five participants completed moderated usability tasks (T1–T5) on laptop, tablet, and mobile devices. Think-aloud comments, task completion, user errors, requests for assistance, and task outcomes were

¹Evaluated build: <http://roommate-haven-dev.eba-2h6p4mxq.ap-southeast-2.elasticbeanstalk.com>.

recorded throughout the sessions. These observations provided additional evidence beyond manual testing. For example, one participant lost context when keyboard focus escaped from a dialog, while another participant found the Discover tabs difficult to navigate using the keyboard and reported that the selected tab relied too heavily on colour alone.

Following the Sprint 2 improvements, the team repeated testing (7 of 7 tests passed) together with testing at 1440×1000 and 390×844 pixels. A WCAG criterion was recorded as *Met* only when both the updated prototype and the supporting regression evidence demonstrated compliance. Criteria that could not yet be fully verified or required implementation beyond the prototype were recorded as *Partially met*.

2.3 WCAG Checklist Results and Actions

Table 2. WCAG 2.0 checklist results, traceability and actions.

SC	Level	Requirement	Traceability	Result	Evidence, consequence and action
1.1.1	A	Non-text Content	US-09, AC-US09-04; US-11	Partially met	Icon controls sampled during the review carried accessible names, and verification cues now pair an icon with status text. The desktop mutual-sharing icon remains the known exception. Audit accessible names across every icon control before claiming conformance.
1.3.1	A	Info and Relationships	NFR-03; US-09, US-10	Partially met	On the sampled components, structure is now programmatic rather than visual: tabs expose <code>role="tab"</code> , <code>tabpanel</code> and <code>aria-controls</code> ; form errors use <code>aria-describedby</code> ; and the blocked-message warning is associated with the composer. Extend the same audit to S-01 and S-03.
1.3.2	A	Meaningful Sequence	NFR-03; S-01 to S-06	Met	Content follows a logical reading and interaction sequence across the evaluated screens. Retain this sequence when responsive layouts change.
1.3.3	A	Sensory Characteristics	NFR-03; T1 to T5	Met	Instructions name controls and actions rather than relying only on position, shape or sound. Review new instructional copy before release.
1.4.1	A	Use of Color	NFR-03; US-11, AC-US11-01	Partially met	Verification cues now pair text with icons and tabs expose <code>aria-selected</code> . However, User B found that the original selected state relied mainly on green text and a thin underline. Retain programmatic state and add a stronger visible non-colour cue such as weight or a selected label.

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Table 2. WCAG 2.0 checklist results continued.

SC	Level	Requirement	Traceability	Result	Evidence, consequence and action
1.4.3	AA	Contrast (Minimum)	NFR-03; S-01 to S-06	Met	The initial white-on-sage controls measured about 3.57:1 and small sage labels about 3.20:1, below the 4.5:1 AA threshold. Sprint 2 changed the palette; checks reported sage 4.95:1, coral 4.91:1, blue 4.56:1 and amber 5.59:1. Recheck after future palette changes.
2.1.1	A	Keyboard	NFR-03; US-11; US-02 and US-10	Partially met	The login dialog, notification drawer and the roster proxy dialog now trap focus, close with Escape and return focus. Tabs support Left/Right, Home and End keys. These fixes address the observed T2 and T5 failures, but a complete keyboard walkthrough of S-01 to S-06 has not been recorded. Add this as a release regression test.
2.1.2	A	No Keyboard Trap	NFR-03; US-09, US-10	Met	Focus containment uses the standard exit method: Escape closes the top-most dialog and restores focus to the opening control, so keyboard users are never held inside a surface. Re-verify whenever a new dialog is added.
2.4.1	A	Bypass Blocks	NFR-03; S-01 to S-06	Met	The original Skip to content link changed the hash but left focus on the page body. The revised link moves focus to <code>#main-content</code> , allowing keyboard and screen-reader users to bypass repeated navigation. Retest whenever routing or landmarks change.
2.4.2	A	Page Titled	NFR-03; S-01 to S-06	Partially met	Each evaluated screen has a clear task heading, but unique browser titles were not separately verified. Add view-specific titles to the next regression pass.
2.4.3	A	Focus Order	NFR-03; US-09, US-10	Met	Opening a dialog or the notification drawer moves focus into it and makes the background inert, and closing returns focus to the trigger. This removed the User A failure in which focus escaped into an <code>aria-hidden</code> drawer. Sample the remaining screens in Iteration 3.
2.4.4	A	Link Purpose (In Context)	NFR-03; T1 to T5	Met	Navigation and task controls state their destination or outcome rather than using ambiguous link text. Recheck new links before release.
2.4.6	AA	Headings and Labels	NFR-03; S-01 to S-06	Partially met	Major headings and form labels describe their content, but icon-only actions need a final label review. Add persistent text where purpose is unclear.

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Table 2. WCAG 2.0 checklist results continued.

SC	Level	Requirement	Traceability	Result	Evidence, consequence and action
2.4.7	AA	Focus Visible	NFR-03; S-01 to S-06	Met	A visible focus indicator was confirmed on the login dialog, skip link, Discover tabs and roster fields during regression capture. Keep the indicator when the palette changes, because contrast and focus styling share the same tokens.
3.1.1	A	Language of Page	NFR-06; US-02	Met	The interface has a consistently identified default language; translated content is assessed separately under SC 3.1.2.
3.1.2	AA	Language of Parts	NFR-06; US-02, AC-US02-02	Partially met	Simplified Chinese rule content now receives <code>lang="zh-CN"</code> , so assistive technology can select appropriate pronunciation. Surrounding headings and actions remain English after Chinese is selected. Translate the complete task flow or mark each remaining language change.
3.2.1	A	On Focus	NFR-03; T1 to T5	Met	Receiving focus alone does not navigate, submit or open a new surface. Retain deliberate activation for future controls.
3.2.2	A	On Input	NFR-06; US-02, AC-US02-02	Met	Changing the rules language or moving between tabs updates content in place without navigating, submitting or opening a new context. Retain this behaviour when translation moves to a service call.
3.2.3	AA	Consistent Navigation	NFR-03; S-01 to S-06	Met	Primary navigation keeps the same relative order throughout the evaluated task flows. Preserve this order when adding destinations.
3.2.4	AA	Consistent Identification	NFR-03; US-02, US-10	Partially met	Repeated actions mostly use consistent terminology, but <i>Household</i> and <i>House rules</i> remain inconsistent. Standardise both labels.
3.3.1	A	Error Identification	NFR-03; US-10, AC-US10-04	Partially met	Verified on the roster proxy dialog: validation applies <code>aria-invalid</code> , links the error through <code>aria-describedby</code> and focuses the first invalid field. The AC-US10-04 rule-proposal form and the login and profile inputs are not yet verified; apply and test the same pattern on all Sprint 1 forms.
3.3.2	A	Labels or Instructions	NFR-03; US-10, AC-US10-04	Partially met	Sampled inputs carry persistent visible labels, and the roster error states the required action rather than only reporting failure. Review the same wording once the rule-proposal form is retested.

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Table 2. WCAG 2.0 checklist results continued.

SC	Level	Requirement	Traceability	Result	Evidence, consequence and action
3.3.3	AA	Error Suggestion	NFR-03; US-10, AC-US10-04	Partially met	Sampled validation messages state how to correct errors, but the pattern is not verified across every Sprint 1 form. Extend the same wording to all forms.
4.1.2	A	Name, Role, Value	US-09, AC-US09-04; US-10	Partially met	Revised tabs expose tab, tabpanel, selected and controlled relationships, and the blocked-message warning is programmatically associated with the composer. The desktop mutual-sharing action still appears icon-only without hover. Add a persistent visible name and verify accessible names for every icon control.

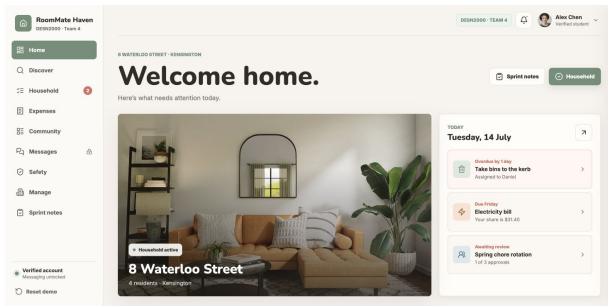
Together, the two tables cover all 38 Level A and Level AA success criteria in the final WCAG 2.0 Recommendation.

Table 3. Checklist items that are not applicable, deferred to the coded solution, or not yet assessed.

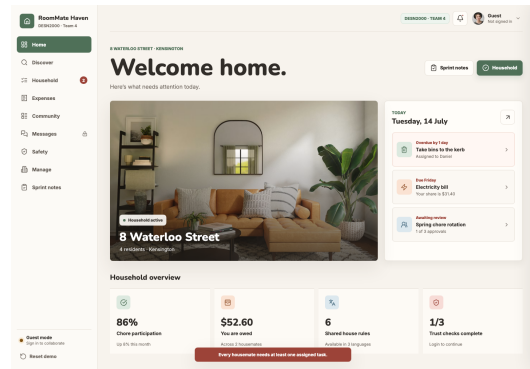
SC	Level	Classification	Reason and planned action
1.2.1 to 1.2.5 (time-based media)	A, AA	Not applicable	The prototype contains no prerecorded or live audio or video. Captions and audio description will be authored alongside any future demonstration media.
1.4.2 Audio Control	A	Not applicable	No audio plays automatically anywhere in the build.
1.4.4 Resize Text	AA	Not yet assessed	Add a 200 per cent zoom and reflow pass to the Iteration 3 regression suite.
1.4.5 Images of Text	AA	Not applicable	The prototype contains no images of text; future interface copy will remain as text.
2.2.1 Timing Adjustable; 2.2.2 Pause, Stop, Hide	A	Final coded solution	Seeded sessions never expire and no content auto-updates. Production sign-in must warn before session expiry and allow the user to extend it.
2.3.1 Three Flashes or Below Threshold	A	Not applicable	No flashing, blinking or auto-playing motion content is present.
2.4.5 Multiple Ways	AA	Final coded solution	The seeded build offers one navigation route per destination. Production will add site search and a rules index.
3.3.4 Error Prevention (Legal, Financial, Data)	AA	Final coded solution	Rule activation and rent payments are currently one-step. The coded product will add a review-and-confirm step and a reversal path before submission.
4.1.1 Parsing	A	Not yet assessed	Add markup validation to the Iteration 3 testing.

2.4 Evaluation Evidence

Figures 1–4 pair each original finding with the revised state, showing how the evidence led to a specific change.

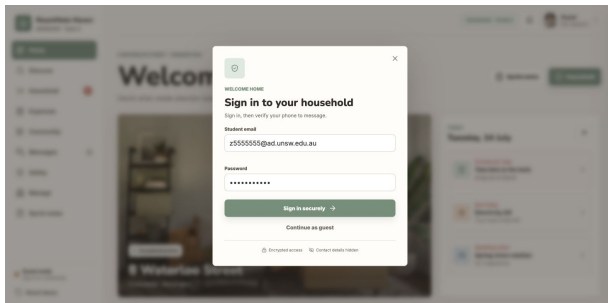


(a) Iteration 1: small controls and status labels failed the AA contrast threshold.

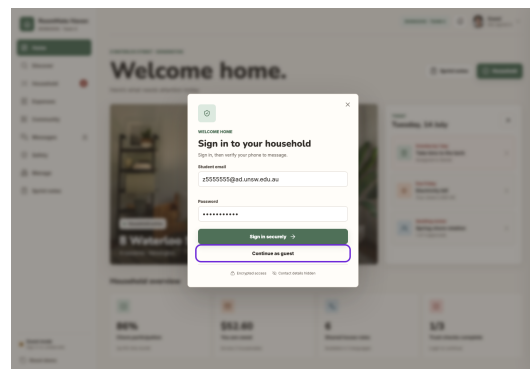


(b) Sprint 2: revised status colours measured at least 4.56:1.

Figure 1. WCAG 1.4.3 contrast evidence before and after palette revision.

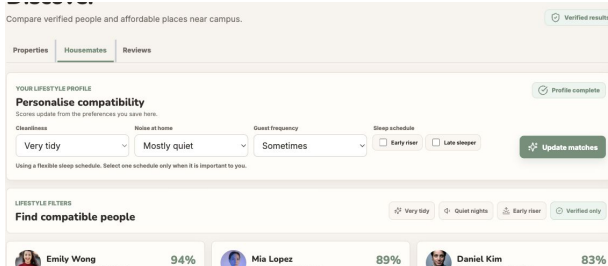


(a) Iteration 1: the dialog remained open after focus moved to a hidden background control.

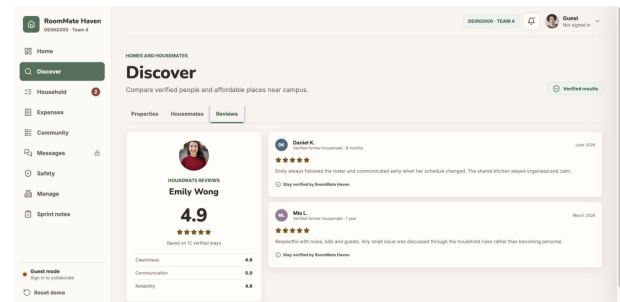


(b) Sprint 2: the visible focus indicator remains within the login dialog; verified the active element.

Figure 2. WCAG 2.1.1 modal keyboard evidence before and after focus management.

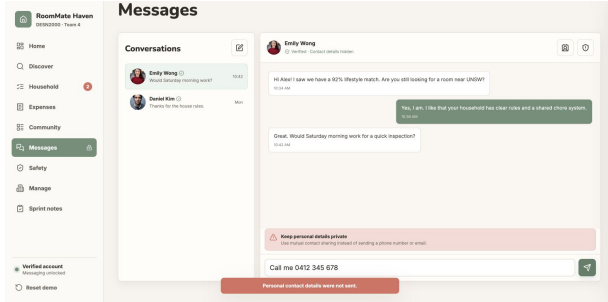


(a) User B/T2: the selected tab relied mainly on colour and did not respond to arrow keys.

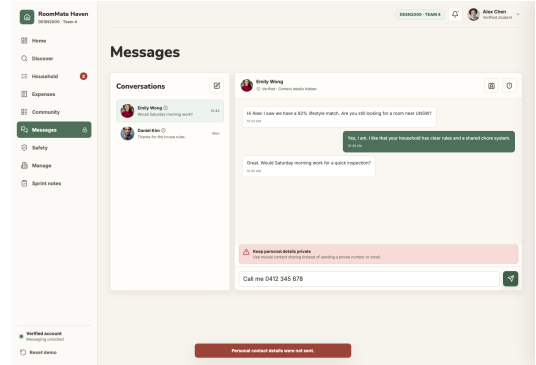


(b) Iteration 2: keyboard navigation exposes focus and the programmatic selected state.

Figure 3. WCAG 1.4.1, 2.1.1 and 4.1.2 evidence for the Discover tabs.



(a) Iteration 1: the blocked-contact warning was visible but had no alert or live-region semantics.



(b) Sprint 2: the warning is associated with the composer and announced as an assertive alert.

Figure 4. Programmatic warning evidence supporting AC-US09-04 and WCAG 4.1.2; live announcement is an additional WCAG 2.1/2.2 improvement.

2.5 Limitations and Next Steps

This evaluation does not establish full conformance. Results marked *Met* are met on the components sampled, not proven across every screen. No participant used a screen reader, switch device or voice control, and seeded session data may hide barriers that production authentication, asynchronous errors and dynamic updates will introduce. The team will next run NVDA with Chrome and VoiceOver with Safari, and repeat the keyboard walkthrough for S-01 to S-06. The remaining interface work is to label the desktop contact-sharing icon, combine approval count and activation condition into one announced status, translate the full household-rule workflow, and apply the roster error pattern to all Sprint 1 forms. Screen-reader and low-vision participants will then validate these changes in practice.

3 Testing Report

3.1 Testing Rationale and Objectives

The usability test assessed whether the prototype allowed participants to complete the core tasks without help. It checked each acceptance criterion and showed where users misunderstood a control, a message, or a change in system state. The team compared observed behaviour with completion data and participant comments. The findings were then used to decide which requirements and screens needed revision before the next iteration.

3.2 Participants and Method

3.2.1 Participants

Five purposively selected participants covered student and working-adult contexts, different rental and shared-living experience, colour-accessibility, mobile-first use and different English-language backgrounds (Table 4). They were recruited through the project team network and screened for these contexts. Participation was voluntary and unpaid, and consent was obtained before each session.

Table 4. Iteration 1 participant profile.

ID	User group	Relevant context
User A	UNSW student	High cleanliness standards and interest in household hygiene.
User B	Domestic UNSW student	Colour vision deficiency.
User C	Employed UNSW graduate living in a shared apartment	English is not the first language.
User D	Newly arrived international student	No Australian rental experience; primarily uses a mobile phone.
User E	Young professional	More than three years of shared-living experience; previously searched for housemates and established common-area rules.

3.2.2 Procedure and Measures

Five participants tested version 1.3 of the HTML prototype across one tablet, two laptops and two mobile devices between 25 and 27 July 2026. The allocation is shown in Table 5.

Table 5. Iteration 1 participant device allocation.

User	Date	Device	Browser	Viewport
User A	25 July 2026	iPad tenth generation	Safari	1180 by 820
User B	25 July 2026	MacBook Air M2	Chrome	1440 by 900
User C	26 July 2026	Dell Inspiron 14	Microsoft Edge	1366 by 768
User D	26 July 2026	iPhone 13	Safari	390 by 844
User E	27 July 2026	Samsung Galaxy S22	Chrome	360 by 800

Each moderated session used the same seeded account and task state. Participants attempted T1 to T5 while thinking aloud. The facilitator gave a hint only after a participant was unable to continue for 30 seconds. The observer recorded final-state completion, time, errors, assistance and interpretation of feedback. Wrong controls, pages or unsupported actions counted as errors; assisted completions were not independent. Task means and individual time-threshold compliance are recorded.

3.3 Iteration 1 Testing

3.3.1 Test Plan and Acceptance Criteria

Iteration 1 retained the five tasks and acceptance-criteria identifiers from Part 2, but expanded coverage from the original mobile viewport to tablet, laptop and mobile contexts. Task outcome and acceptance-criteria status were assessed separately. A task was *Completed* only when at least four of five participants independently met every behaviour, interpretation and time condition in its measurable threshold. It was *Partially completed* when participants reached some or all of the workflow but one or more conditions were failed or not separately tested. Each acceptance criterion was classified as Pass, Partial, Fail or Not separately tested. Partial means that some required behaviour was demonstrated but the full measurable threshold was not met; task completion alone did not override a criterion-level result.

Table 6. Iteration 1 test plan and acceptance criteria coverage.

Test	User task	Traceability	Acceptance criteria	Measurable success threshold
T1	Narrow properties using combined search criteria	US-13; S-01	AC-US13-01 to AC-US13-04 and AC-US13-06 (implemented)	At least 4/5 independently complete the filtered search, explain affordability, recover from no results and use a custom address within three minutes.
T2	Recognise verified housemate profiles	US-11; S-02	AC-US11-01 (simulated) and AC-US11-02 (implemented)	At least 4/5 independently find the verified-only control, identify its cues and explain that verification is not a guarantee within 90 seconds.
T3	Build a profile and use recalculated compatibility	US-12; S-03	AC-US12-01, AC-US12-02 and AC-US12-04 to AC-US12-06 (implemented)	At least 4/5 independently complete the profile, explain the top match and recover from validation and zero-result states within three minutes.
T4	Complete phone verification and message safely	US-09; S-04	AC-US09-01 (simulated) and AC-US09-04 (implemented)	At least 4/5 independently pass verification and message within three minutes; contact details remain blocked with a clear explanation.
T5	Propose, review and activate a house rule	US-02, US-10; S-05, S-06	AC-US02-01 to AC-US02-03, AC-US10-01, AC-US10-02 and AC-US10-04 (implemented)	At least 4/5 independently view, propose and activate a rule, including language, validation and approval states, within four minutes.

3.3.2 Testing Outcomes

Table 7. Iteration 1 outcome summary.

Test	Task outcome	Quantitative outcome	Acceptance-criteria assessment and consequence
T1	Partially completed	5/5 reached the final state; 4/5 independently; mean 2 min 8 sec; 2 errors; 1 assisted	AC-US13-01 to AC-US13-03: Pass. AC-US13-04: Fail because participants did not reliably understand how to recover from the empty state. All five participants completed T1 within three minutes, and four did so independently while one required a facilitator hint.
T2	Partially completed	5/5 reached the final state independently; mean 1 min 2 sec; 1 error; no assistance	Simulated AC-US11-01: Pass. Implemented AC-US11-02: Pass. Individual compliance with the 90-second limit was not recorded, so the composite task threshold was not fully verified. Two users found the explanation too small; enlarge it and retain non-colour cues.
T3	Partially completed	4/5 reached the final state; 3/5 independently; mean 2 min 44 sec; 3 errors; 1 assisted	AC-US12-01: Pass. AC-US12-02: Partial because the score source and limitations were not explained accurately enough. AC-US12-04 to AC-US12-06: Not separately tested. Enlarge the factor explanation and show why the score changed.
T4	Partially completed	5/5 reached the final state; 4/5 independently; mean 2 min 18 sec; 2 errors; 1 assisted	Simulated AC-US09-01: Pass. AC-US09-04: Fail. Contact details were blocked, but the criterion's clear explanation and safe next step were not understood. Link the warning to the contact-sharing action.
T5	Partially completed	4/5 reached the final state; 2/5 independently; mean 3 min 11 sec; 4 errors; 2 assisted	AC-US02-01: Not separately tested. AC-US02-02: Pass. AC-US02-03: Not separately tested. AC-US10-01: Pass. AC-US10-02: Fail because participants could not reliably connect approval progress with activation. AC-US10-04: Not separately tested. Combine and emphasise approval and activation status.

All five reached the final state in T1, T2 and T4, but task-level completion did not establish that every included acceptance criterion had passed. T2 provided the strongest result because it required

no assistance, while T3 and T5 had the lowest independent completion and most errors. Problems clustered around post-action feedback: empty-search recovery, score meaning, contact blocking and rule approval. Iteration 2 will keep participant-level times and separate results for every criterion.

3.3.3 User Feedback

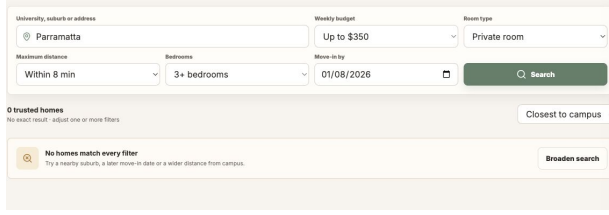
Table 8 reports the seven accepted participant-level findings. Device and viewport allocations are given in Table 5. The statements paraphrase observation notes rather than claiming verbatim participant quotations.

Table 8. Participant-level accessibility and device-specific findings.

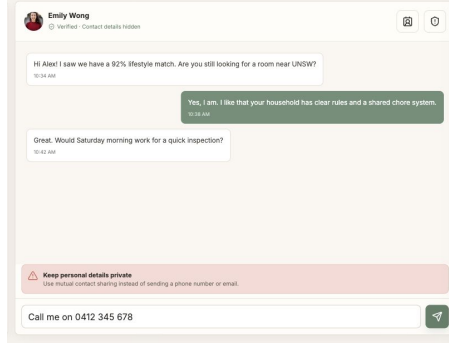
User	Task	Observed issue	Effect on task
A	T5	Keyboard focus moved outside the Create roster dialog. Escape did not close it, and focus was not returned to the opening control after closing.	The participant lost the interaction context and required assistance.
B	T2	The Discover tabs did not respond to the arrow keys. Their selected state also relied mainly on colour and a thin underline.	The participant completed the task with a pointer, but keyboard navigation and the selected state were unclear.
C	T4	The privacy warning appeared after contact details were entered, but its relationship to the icon-only contact-sharing action was unclear.	The participant paused for more than 30 seconds and continued after one facilitator hint.
C	T5	Selecting Chinese translated rule content while surrounding headings, labels and actions remained in English.	The mixed-language state reduced clarity and made the language setting appear incomplete.
D	T4	At 390 by 844 pixels, the conversation list and New message control were hidden with no visible mobile replacement.	This restricted mobile conversation management because the participant could not select Daniel or begin a new conversation.
E	T5	Selecting Save draft produced no visible confirmation, navigation or control-state change.	The participant selected the control twice and assumed it was not working.
E	T5	Approval progress was displayed separately from the activation condition and was not sufficiently prominent.	The participant requested clarification and did not complete the task independently.

Supporting interface states appear in Appendix A.

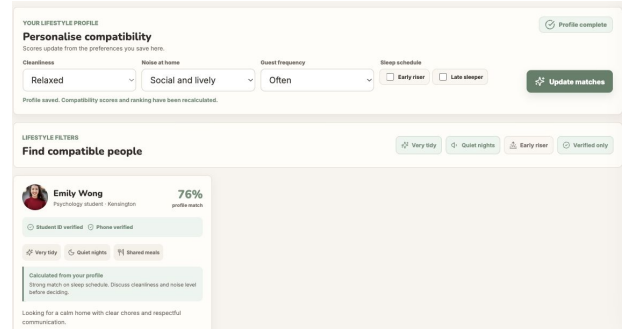
Figure 5 consolidates the original task-level post-session recreations captured from the same Sprint 1 HTML build at a 1440 by 900 viewport on 27 July 2026. They document the interface states before Iteration 2 refinement; the behavioural evidence is reported in Table 7.



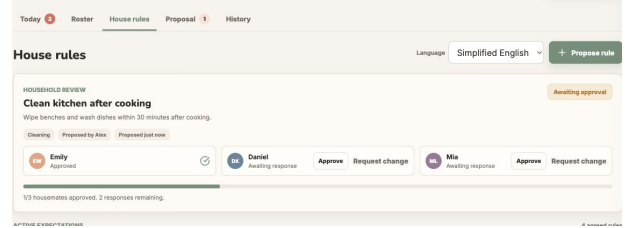
(a) T1, AC-US13-04: the secondary “Broaden search” action supported a more prominent reset action.



(c) T4, AC-US09-04: the warning and icon-only sharing control were separated, making the next step less apparent.



(b) T2–T3, AC-US11-02 and AC-US12-02: small trust and score explanations supported larger text and non-colour cues.



(d) T5, AC-US10-02: separated approval and activation indicators supported a stronger combined status message.

Figure 5. Compact Sprint 1 task-level baseline evidence before Iteration 2 refinement.

3.4 Changes after Sprint 1

The team changed states that failed, remained partial or blocked keyboard and mobile use in Iteration 1. Table 9 links each design change to its evidence and corresponding Jira user-story issue. These links identify the maintained story records.

Table 9. Design changes made after Sprint 1.

Task	Design modification	Evidence	Updated Jira link
T1	The empty state places Broaden search beside guidance that explains how to recover.	Figure 6a	ST4-43
T2	Verification cues pair an icon with readable status text.	Figure 6b	ST4-38
T2	Tabs expose their selected state and respond to the arrow, Home and End keys.	Figure 7	ST4-38
T3	Each match card presents a summary of matching preferences and differences.	Figure 6c	ST4-39
T4	The message warning blocks the content, explains mutual sharing and announces the result through an alert.	Figure 8a	ST4-35
T4	The mobile message view keeps a conversation selector and a New control.	Figure 8b	ST4-35
T5	Open dialogs keep keyboard focus, close with Escape and return focus to the opening control.	Figure 9a	ST4-36
T5	Roster validation links the error to the invalid field and moves focus to it.	Figure 9a	ST4-36
T5	Chinese rule content exposes the zh-CN language code to assistive technology.	Figure 9b	ST4-24
T5	Saving a draft updates the page status and shows a confirmation.	Figure 9c	ST4-36

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Table 9. Design changes made after Sprint 1 continued.

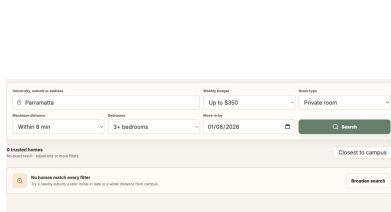
Task	Design modification	Evidence	Updated Jira link
T5	The final acknowledgement moves the rule to the active list and adds an activation timestamp.	Table 18	ST4-36

3.5 Iteration 2 Test

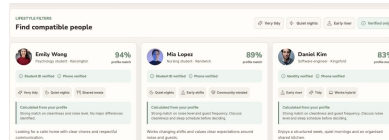
From 4 to 7 August 2026, the same five participants retested version 1.4 on the devices and viewports recorded in Appendix B. Their prior exposure may have shortened completion times. Table 18 records the participant results, while Figures 6 to 9 show the tested interface states.

Table 10. Iteration 2 criterion-level retest plan.

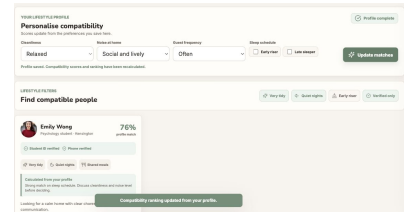
Task	Retested AC	Test case	Success threshold	Recorded measures
T1	AC-US13-04	Starting from a no-results state, independently change the search so that at least one valid result is displayed.	At least 4/5 participants recover independently within 60 seconds, without a facilitator prompt.	Completion time, errors, assistance and interpretation of the recovery guidance.
T3	AC-US12-02	Inspect a compatibility result and identify its match and difference summary.	At least 4/5 independently identify one match and one difference.	Completion time, errors, hints and independence.
T4	AC-US09-04	Attempt to send a message containing personal contact details, then explain whether the details were sent and identify the safe next step.	At least 4/5 independently confirm that the details remain unsent and correctly explain the mutual-sharing condition and next action.	Completion, explanation accuracy, errors and assistance.
T5	AC-US10-02	Complete the final acknowledgement, then confirm that the rule becomes Active, enters the rules list and shows a timestamp.	At least 4/5 independently complete the acknowledgement and find the active rule and timestamp.	Completion time, errors, hints and independence.



(a) T1 shows recovery guidance beside the empty result.



(b) T2 shows text and icons for verification.



(c) T3 presents the match and difference summary.

Figure 6. Iteration 2 implementation evidence for T1 to T3.

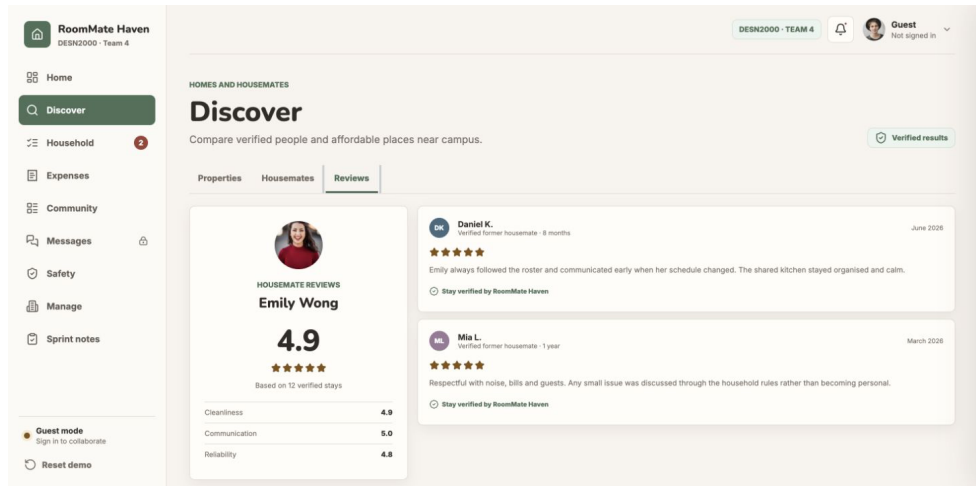
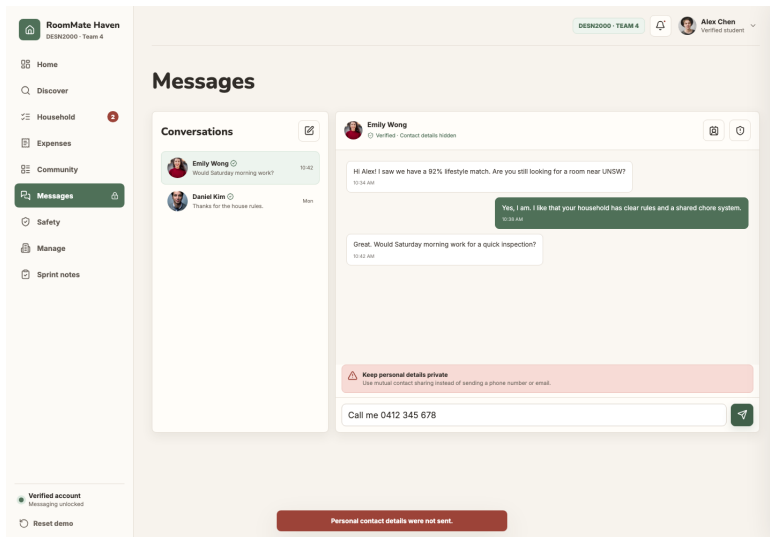
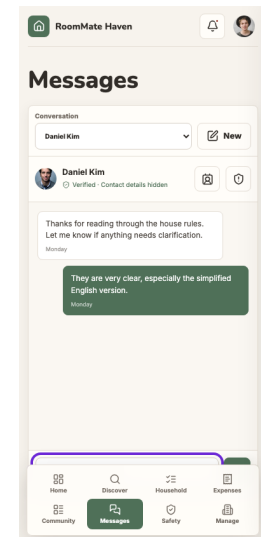


Figure 7. T2 tab focus and selected state after keyboard navigation.

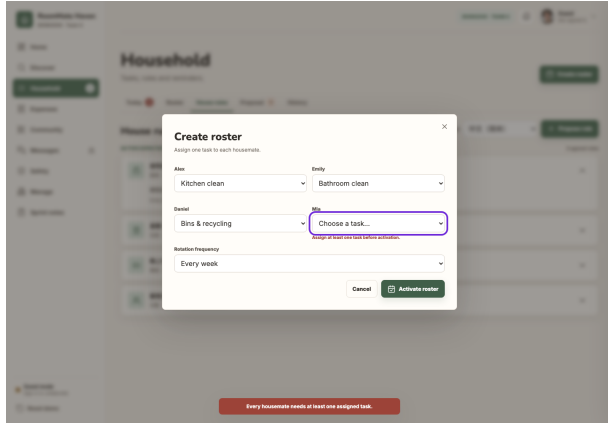


(a) T4 blocks a phone number and shows the safer next step.

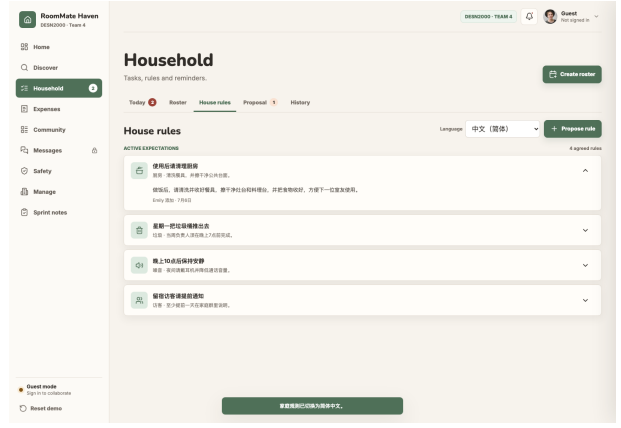


(b) T4 keeps conversation controls at the mobile viewport.

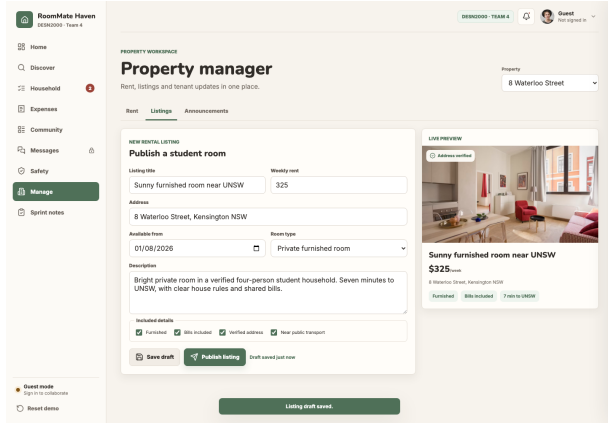
Figure 8. Iteration 2 implementation evidence for T4.



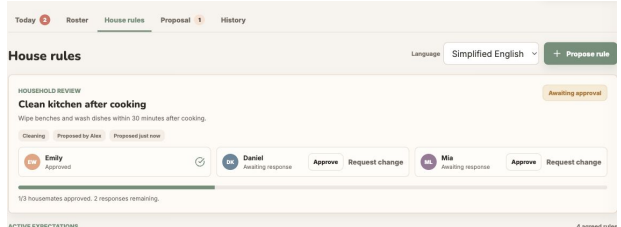
(a) T5 keeps focus on the invalid roster field.



(b) T5 displays Chinese rules with matching language metadata.



(c) The supplementary draft check shows a saved state.



(d) The proposal state precedes the final acknowledgement.

Figure 9. Iteration 2 implementation evidence for T5.

Table 11. Iteration 2 criterion-level retest results.

Task	Ind.	Assist.	Time	Err.	Hint	Observed result	Status
T1	5/5	0/5	44 seconds median	3	0	5/5 recovered independently within 60 seconds.	Pass
T3	3/5	2/5	70 seconds median	4	2	3/5 found both parts of the summary without help.	Partial*
T4	5/5	0/5	52 seconds median	0	0	5/5 understood the block and the next step.	Pass
T5	4/5	1/5	75 seconds median	3	1	4/5 completed the acknowledgement and found the timestamp without help.	Pass

*Partial indicates that three of five participants met the complete criterion threshold.

Table 12. Iteration 2 results for the modified Sprint 1 tasks.

Task	Retest outcome and resolved problem	Acceptance criteria result	New issue	Remaining problem
T1	Completed for the targeted criterion. All five recovered without help within 60 seconds.	AC-US13-04 changed from Fail to Pass. Other T1 criteria were not separately retested.	Yes.	The recovery control took longer to find on both mobile viewports.
T2	Supplementary regression check. Text supports each verification cue. Keyboard users can identify and change tabs.	AC-US11-02 remained Pass from Iteration 1; no other T2 criterion was separately retested.	No.	Profile identity and phone verification still use similar visual cues.
T3	Partially completed for the targeted criterion. Three participants found both summary sections without help, while two needed a hint.	AC-US12-02 remained Partial. Other T3 criteria were not separately retested.	Yes.	Both mobile participants missed the difference section below the fold.
T4	Completed for the targeted criterion. All five understood why the details were blocked and identified the next step without help.	AC-US09-04 changed from Fail to Pass. Other T4 criteria were not separately retested.	No.	The desktop sharing action still relies on an icon, so its purpose is less clear without hover.
T5	Completed for the targeted criterion. Four participants completed the acknowledgement and found the timestamp without help.	AC-US10-02 changed from Fail to Pass. Other T5 criteria were supplementary checks or were not separately retested.	Yes.	The activation timestamp remains less prominent on mobile.

3.6 Open Issues and Impact

Table 13. Open issues and participant counts.

Issue	Iteration 1 n/5	Iteration 2 n/5	Impact	Next action
Search recovery control	4/5 exceeded 60 seconds	2/5 hesitated on mobile	Mobile recovery may take longer.	Increase the prominence of the recovery control.
Difference summary visibility	4/5 needed a hint	2/5 needed a hint	A match may be judged without its differences.	Place both summary sections before secondary content.
Approval and activation status	1/5 documented	1/5 needed a hint	An active rule may remain unclear.	Place the Active label beside its timestamp.

4 Final Prototype

4.1 Overview and Fidelity

The final RoomMate Haven prototype is available as a deployed browser build: <http://roommate-haven-dev.eba-2h6p4mxq.ap-southeast-2.elasticbeanstalk.com>. A matching 60-second walkthrough is stored in the repository at `media/RoomMate-Haven-Prototype-Demo-60s.mp4`. The final prototype retains the six-screen Sprint 1 journey: property search (S-01), verified

housemate discovery (S-02), lifestyle compatibility (S-03), private messaging (S-04), household rules (S-05), and rule approval (S-06).

The build is a high-fidelity, seeded HTML prototype rather than a production service. It provides realistic interaction, presentation and content for the assessed journey while keeping test conditions repeatable. The evidence in Table 14 is limited to behaviour recorded in the repository and does not imply production identity verification, encrypted messaging, server-side persistence or complete WCAG conformance.

Table 14. Final prototype fidelity and repository evidence.

Fidelity dimension	Implemented evidence	Scope boundary
Interactivity	Filters produce result and no-result states; saved lifestyle choices recalculate rankings; contact details are blocked before sending; dialogs manage focus and validation; tabs respond to keyboard input; and saved actions provide confirmation. Figures 6, 7, 8 and 9 show the revised states.	Interactions use seeded browser data. The Accessibility Report records 7/7 checks.
Visual	Responsive desktop and mobile layouts use a consistent warm-white, sage and peach system, repeated cards and controls, readable type, visible focus and status styling. The revised status colours measured between 4.56:1 and 5.59:1 in the recorded contrast checks (Figure 1).	Visual evidence covers the tested 1440 by 1000 and 390 by 844 layouts; it is not evidence for every possible viewport.
Content	Seeded properties, profiles, compatibility explanations, conversations, verification states, household members and rule proposals provide realistic labels, values, warnings and state changes for T1–T5.	Seeded identities and household data are demonstration content and reset with the prototype; they are not live records.

4.2 Design Principle Evidence Matrix

Table 15 applies the information-architecture, visual- hierarchy and interface principles documented in the Week 7 design notes to observable prototype states. It contains more than the rubric minimum of five information architecture principles, three visual hierarchy principles and five UI principles.

Table 15. Design principles implemented in the final prototype.

Category	Principle	Prototype implementation	Evidence
Information architecture	Objects	Properties, housemate profiles, conversations and household rules are treated as distinct objects with their own attributes, statuses and actions.	Figures 6–9
Information architecture	Choices	Each surface prioritises a small task-relevant set of controls, such as property filters, lifestyle chips or approval actions, rather than exposing the full product backlog at once.	Figures 6 and 9
Information architecture	Progressive disclosure	Match cards first show score, source and one discussion point; validation, warnings and recovery guidance appear when the corresponding state is reached.	Figures 6 and 8
Information architecture	Exemplars	Seeded listings, profiles, messages and proposals demonstrate the expected content and status format before a participant creates or changes data.	Figures 6–9

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Table 15. Design principles implemented in the final prototype continued.

Category	Principle	Prototype implementation	Evidence
Information architecture	Front doors	Direct routes such as #discover/properties , #discover/housemates , #messages and #household/rules open a labelled task surface with a visible current state.	Figure 7; deployed prototype links for S-01–S-06
Information architecture	Multiple classification	Properties can be narrowed by location, budget, room type, distance, bedrooms and date, while housemates can be classified by lifestyle and verification attributes.	Figure 6
Visual hierarchy	Scale and typography	Large page headings establish context, card titles identify objects, and smaller labels and metadata provide supporting detail in a consistent reading order.	Figures 6 and 8
Visual hierarchy	Colour and contrast	Sage identifies primary and trusted states, coral identifies blocked or urgent states, and text/icons supplement colour. Recorded status colours meet the sampled AA contrast threshold.	Figures 1 and 4
Visual hierarchy	Proximity and spacing	Filters are grouped above results, conversation controls stay beside their message context, and each member’s approval state stays within the same rule card.	Figures 6, 8 and 9
Visual hierarchy	Repetition and alignment	Repeated cards, status chips, aligned form controls and recurring navigation patterns let users scan comparable content across desktop and mobile.	Figures 6–9
UI	Visibility of system status	Compatibility recalculation, draft saving, contact blocking and approval progress all produce visible state or confirmation feedback.	Figures 6, 8 and 9
UI	User control and freedom	Users can cancel dialogs, close them with Escape, return focus to the opening control, change filters and reset the demonstration state.	Figure 2
UI	Consistency and standards	Shared buttons, fields, cards, dialogs and tab patterns use consistent placement and interaction behaviour across the tested tasks.	Figures 7 and 9
UI	Error prevention	Contact details are stopped before message submission, and incomplete roster data prevents activation while identifying the invalid field.	Figures 8 and 9
UI	Recognition over recall	Persistent field labels, text-plus-icon verification cues, selected tabs and visible household-member statuses reduce the need to remember earlier screens.	Figures 3 and 9
UI	Error recovery	The empty search state explains which inputs can change and offers a Broaden search action; validation moves focus to the first field requiring correction.	Figures 6 and 9

4.3 Accessibility Integration

Table 16 is a concise integration summary, not a second accessibility audit. It includes only requirements recorded as *Met*. The evaluation method, participant evidence, partially met criteria, deferred items and limitations remain in Section 2, especially Table 2.

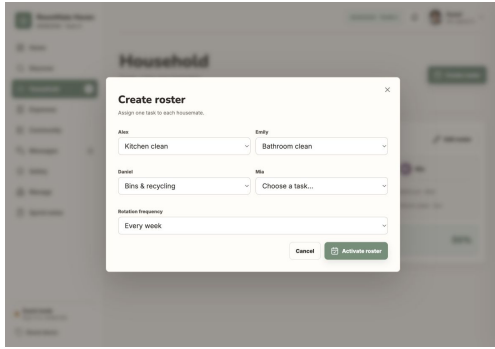
Table 16. Implemented WCAG requirements in the final prototype.

WCAG SC	Requirement	Implemented evidence	Detailed evidence
1.4.3	Contrast (Minimum)	Revised sage, coral, blue and amber status colours measured at least 4.56:1 in the recorded checks.	Figure 1
2.1.2	No Keyboard Trap	Escape closes the top-most dialog and restores focus to its trigger.	Figure 2
2.4.1	Bypass Blocks	The skip link moves keyboard focus to <code>#main-content</code> .	Table 2
2.4.3	Focus Order	Dialogs and the notification drawer receive focus, make the background inert and return focus when closed.	Figure 2
2.4.7	Focus Visible	Visible focus was captured on the login dialog, skip link, Discover tabs and roster fields.	Figures 2 and 3

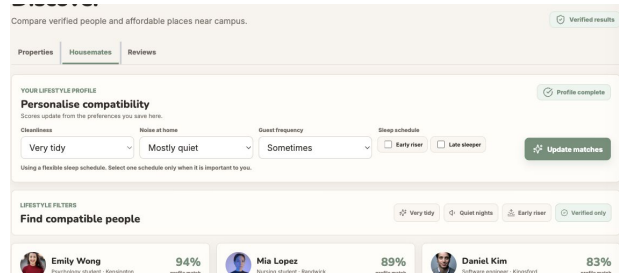
4.4 Evidence Boundaries

The matrices describe implemented prototype evidence, not unqualified success across the whole product. The second iteration recorded Partial for AC-US12-02, and Pass for AC-US13-04, AC-US09-04 and AC-US10-02. The open participant findings appear in Section 3.6. The interface review also retained the Household and House rules label mismatch, an icon-only desktop sharing action and mixed-language controls.

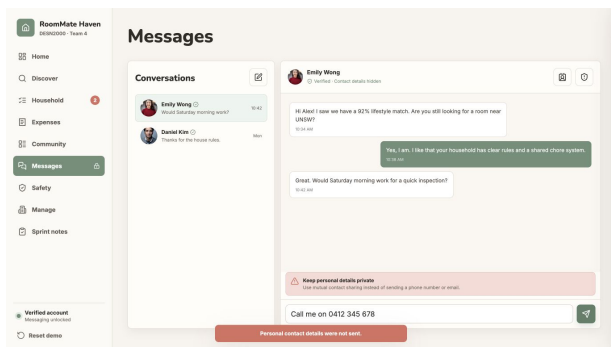
A Iteration 1 Participant Evidence



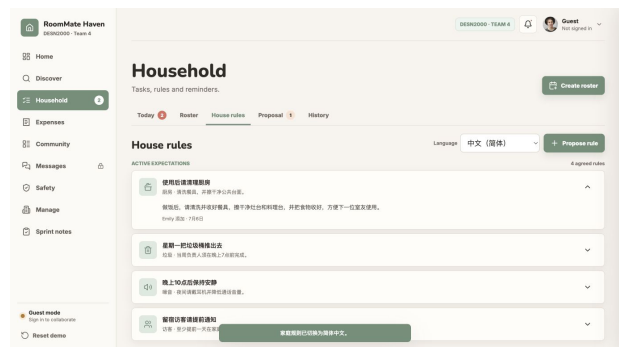
(a) User A T5. At 1180 by 820 pixels, keyboard focus left the open dialog and Escape did not close it. Focus was not restored to the trigger, so User A needed help.



(b) User B T2. At 1440 by 900 pixels, the tabs ignored the arrow keys and showed selection mainly through green text and a thin underline. User B used a pointer to finish.

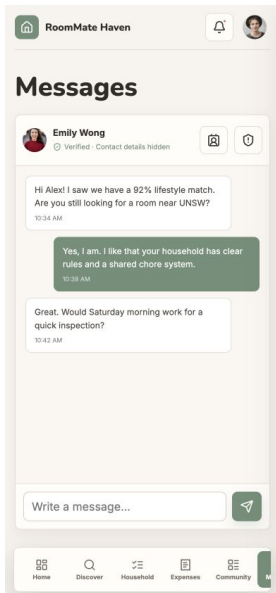


(c) User C T4. At 1366 by 768 pixels, the warning did not clarify how to use the separate contact sharing icon. User C paused for more than 30 seconds and needed one hint.

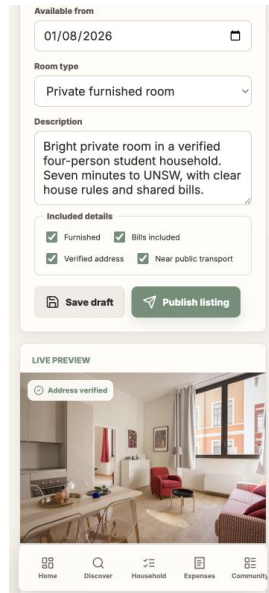


(d) User C T5. At 1366 by 768 pixels, selecting Simplified Chinese translated the rules but left the surrounding interface in English. User C found the setting incomplete.

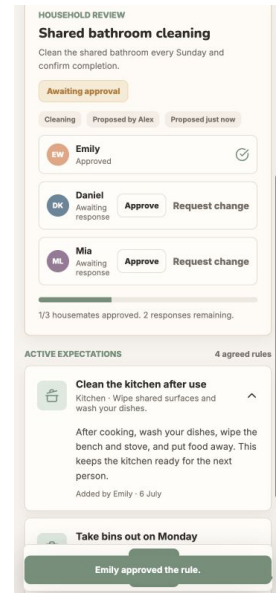
Figure 10. Participant evidence from tablet and laptop sessions.



(a) User D T4. At 390 by 844 pixels, the screen hid the conversation list and New message control without a replacement. User D could not select Daniel or start a chat.



(b) User E T5. At 360 by 800 pixels, Save draft produced no visible response. User E selected it twice and assumed it had failed.



(c) User E T5. At 360 by 800 pixels, the approval count appeared apart from the activation condition. User E needed clarification and did not finish independently.

Figure 11. Participant evidence from mobile sessions.

B Anonymised Iteration 2 Participant Records

All records used prototype version 1.4 with the device allocation in Table 17, and aggregate AC status follows the thresholds in Table 10.

Table 17. Iteration 2 participant device allocation.

Participant	Iteration	Version	Device	Browser	Viewport
A	2	1.4	iPad tenth generation	Safari	1180 by 820
B	2	1.4	MacBook Air M2	Chrome	1440 by 900
C	2	1.4	Dell Inspiron 14	Microsoft Edge	1366 by 768
D	2	1.4	iPhone 13	Safari	390 by 844
E	2	1.4	Samsung Galaxy S22	Chrome	360 by 800

Table 18. Iteration 2 participant observation records.

User	Date	Test and AC	Time	Err.	Hint	Ind.	Result	Obs. ID	Interpretation
A	4 Aug 2026	T1 / AC-US13-04	42 s	0	0	Yes	Pass	OBS-A-T1	Recognised the no results state and changed the search query.
B	5 Aug 2026	T1 / AC-US13-04	35 s	0	0	Yes	Pass	OBS-B-T1	Recovered from the no results state by changing the search terms.
C	6 Aug 2026	T1 / AC-US13-04	58 s	1	0	Yes	Pass	OBS-C-T1	Repeated the same search once, then changed the query without help.
D	7 Aug 2026	T1 / AC-US13-04	58 s	1	0	Yes	Pass	OBS-D-T1	Hesitated because the mobile recovery action was less prominent.
E	6 Aug 2026	T1 / AC-US13-04	44 s	1	0	Yes	Pass	OBS-E-T1	Recovered without help after scanning for the recovery control.
A	5 Aug 2026	T3 / AC-US12-02	55 s	0	0	Yes	Pass	OBS-A-T3	Found and understood the match and difference summary.
B	7 Aug 2026	T3 / AC-US12-02	41 s	0	0	Yes	Pass	OBS-B-T3	Identified both matches and differences.
C	4 Aug 2026	T3 / AC-US12-02	70 s	1	0	Yes	Pass	OBS-C-T3	Initially missed the difference section, then found it after scrolling.
D	5 Aug 2026	T3 / AC-US12-02	92 s	1	1	No	Fail	OBS-D-T3	Found the difference summary after a prompt to scroll.
E	4 Aug 2026	T3 / AC-US12-02	105 s	2	1	No	Fail	OBS-E-T3	Found the match summary but needed help to find the difference summary.
A	4 Aug 2026	T4 / AC-US09-04	48 s	0	0	Yes	Pass	OBS-A-T4	Saw that contact information was blocked and understood the next step.
B	6 Aug 2026	T4 / AC-US09-04	39 s	0	0	Yes	Pass	OBS-B-T4	Understood the block and followed the suggested next action.
C	7 Aug 2026	T4 / AC-US09-04	52 s	0	0	Yes	Pass	OBS-C-T4	Understood the blocking message and identified the next step.
D	4 Aug 2026	T4 / AC-US09-04	64 s	0	0	Yes	Pass	OBS-D-T4	Understood both the block and the next step.
E	5 Aug 2026	T4 / AC-US09-04	72 s	0	0	Yes	Pass	OBS-E-T4	Understood why contact details were blocked and what to do next.
A	6 Aug 2026	T5 / AC-US10-02	62 s	0	0	Yes	Pass	OBS-A-T5	Completed the final confirmation and found the activation timestamp.
B	5 Aug 2026	T5 / AC-US10-02	49 s	0	0	Yes	Pass	OBS-B-T5	Found the active rule and timestamp after confirmation.
C	6 Aug 2026	T5 / AC-US10-02	75 s	1	0	Yes	Pass	OBS-C-T5	Activated the rule but took longer to locate the timestamp.
D	7 Aug 2026	T5 / AC-US10-02	80 s	1	0	Yes	Pass	OBS-D-T5	Completed confirmation but did not notice the timestamp at once on mobile.
E	7 Aug 2026	T5 / AC-US10-02	98 s	1	1	No	Fail	OBS-E-T5	Activated the rule but needed a hint to find the timestamp.

References

- [1] World Wide Web Consortium (W3C). *Web Content Accessibility Guidelines (WCAG) 2.2*. W3C Recommendation, 12 December 2024. <https://www.w3.org/TR/WCAG22/> (accessed 12 August 2026).
- [2] World Wide Web Consortium (W3C). *Web Content Accessibility Guidelines (WCAG) 2.0*. W3C Recommendation, 11 December 2008. <https://www.w3.org/TR/2008/REC-WCAG20-20081211/> (accessed 12 August 2026).
- [3] World Wide Web Consortium (W3C), Web Accessibility Initiative (WAI). *Template for Accessibility Evaluation Reports*. Updated April 2024. <https://www.w3.org/WAI/test-evaluate/report-template/> (accessed 12 August 2026).
- [4] The A11Y Project. *Checklist*. <https://www.a11yproject.com/checklist/> (accessed 12 August 2026).